

Functional & Performance Testing Template

Model Performance Test

Date	04 Jul 2026
Team ID	
Project Name	Project: Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
FT-01	User Registration Validation	Enter valid and invalid registration details (name, email, password, location).	Valid users can register successfully; invalid inputs display appropriate error messages.	As Expected	Pass
FT-02	User Login Validation	Log in using valid and invalid credentials.	Valid users are redirected to the dashboard; invalid credentials show an error message.	As Expected	Pass
FT-03	Weather Data Retrieval	Connect to the weather API and request current weather information.	Weather data (rainfall, temperature, humidity, etc.) is retrieved successfully.	As Expected	Pass
FT-04	River Level Data Collection	Fetch river water level data from sensors or APIs.	Latest river level data is displayed correctly.	As Expected	Pass
FT-05	Machine Learning Flood Prediction	Provide historical and real-time environmental data and execute the prediction model.	System accurately predicts flood risk (Low, Medium, High, Critical).	As Expected	Pass
FT-06	Flood Alert Notification	Trigger a flood-risk event and verify alert delivery.	SMS, email, and mobile notifications are sent successfully.	As Expected	Pass

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass/Fail
FT-07	Interactive Flood Risk Map	Select a location and view the flood risk map.	Flood-prone areas are displayed accurately on the map.	As Expected	Pass
FT-08	Report Generation	Generate and download a flood prediction report.	Report is generated successfully in PDF format.	As Expected	Pass
PT-01	Dashboard Response Time	Load the application dashboard under normal conditions.	Dashboard loads within 3 seconds .	2.4 seconds	Pass
PT-02	Prediction Response Time	Submit prediction requests simultaneously from multiple users.	Flood prediction results are generated within 5 seconds .	4.1 seconds	Pass
PT-03	Concurrent User Load Test	Simulate 100 concurrent users accessing the system.	Application remains stable without crashes or significant delays.	Stable Performance	Pass
PT-04	Database Performance Test	Retrieve large historical flood datasets from the database.	Data retrieval completes within acceptable response time (<2 seconds).	1.8 seconds	Pass
PT-05	API Performance Test	Send multiple requests to weather and river-level APIs simultaneously.	APIs respond successfully without timeout errors.	As Expected	Pass
ST-01	User Authentication Security	Attempt login with invalid credentials multiple times.	Unauthorized access is blocked and error message displayed.	As Expected	Pass
ST-02	Data Encryption	Verify that user credentials and API communications use HTTPS encryption.	All sensitive data is securely encrypted during transmission.	As Expected	Pass
UT-01	Mobile Device Compatibility	Test application on Android and iOS devices.	Application functions correctly across supported mobile devices.	As Expected	Pass
UT-02	Web Browser Compatibility	Test application on Chrome, Firefox, Edge, and Safari browsers.	Application works consistently across all supported browsers.	As Expected	Pass